

Data Package & Analysis Guide

A Public-Data Review-Priority Framework for Medicaid Residential Care Oversight: Evidence from Washington State Adult Family Homes

Author: Mohamed Noor Hussein **Prepared:** June 27, 2026 **Scope:** Every dataset used in the manuscript, what it contains, where it came from, and exactly what analysis was performed on it.

How this package is organized

This archive contains two evidence bases:

1. **Washington State AFH/HCBS evidence base** (`wa_data/`) — the primary empirical pilot covering King, Pierce, and Spokane counties.
2. **Minnesota Medicaid evidence base** (`mn_data/`) — the enforcement-record anchor used as a structural analogue.

A reproducible-analysis note for each file is given below. All facility-level Washington data was assembled from **public** sources only (DSHS license roster, DSHS RCS public-document portal, WA HCA exclusion list, U.S. Census ACS). No protected health information or non-public claims data was used.

Evidentiary discipline. No Washington provider in these files is alleged to have committed fraud. Facility-level indicators are *review-priority* signals, not findings of wrongdoing. All Minnesota cases are described as **alleged / charged** pending judicial resolution.

PART A — WASHINGTON DATASETS (`wa_data/`)

A1. `WA_AFH_Facility_Master.csv` — Statewide AFH roster (raw)

- **Rows x cols:** 6,075 x 24 (one row per active AFH license, statewide, deduplicated by license number)
- **Source:** DSHS-published **ArcGIS feature service** "Long Term Care – Residential Care" (the nightly extract behind the official AFH locator), layer `Long_Term_Care_Residential_Care_view/FeatureServer/1`. Accessed 2026-06-26.
- **Key fields:** `License_Number`, `Facility_Name`, `License_Status`, `Physical_Address`, `City`, `ZIP`, `County`, `Phone`, `Licensed_Capacity`, `Specialty`, `Contract` (Medicaid contract designation), `Latitude/Longitude`, `SourceURL`, `DateAccessed`.
- **Analysis performed:** Paginated pull of all features (2,000/page), dedupe by license number, normalization of county/city/phone fields. This is the statewide universe from which the three-county pilot is drawn. Also used to derive the **current 2026 roster count (6,075 active AFHs; 35,305 licensed beds)** that anchors the growth series.

A2. `WA_AFH_3County_Master.csv` — Three-county pilot subset

- **Rows x cols:** 3,457 x 25
- **Source:** Filtered subset of A1 to `County ∈ {King, Pierce, Spokane}`.
- **Analysis performed:** County filter + `Medicaid_Contract_Flag` derivation from the DSHS contract field. This 3,457-facility file is the **denominator** for every rate in the manuscript. County split: **King 1,795 / Pierce 1,052 / Spokane 610**. `Medicaid_Contract_Flag` is a presence proxy — any non-empty contract-field value maps to "Yes (has DSHS contract)"; it does not verify contract terms, scope, or active status, only that the public roster's contract field was non-blank.

A3. `WA_AFH_3County_Reports.csv` — Per-facility public-document tallies (scrape)

- **Rows x cols:** 3,457 x 18
- **Source:** DSHS **RCS public-documents portal**, `fortress.wa.gov/dshs/adsaapps/lookup/AFHForms.aspx?lic=<license>`, one request per license (public records per RCW 70.128.280). Accessed 2026-06-26.
- **Analysis performed:** Threaded scrape of each facility's document folder; document URLs classified by folder token into **inspections / investigations / enforcement / limitations / other**. Enforcement subtypes

decoded from filenames: CF = civil fine, SP = stop placement, Cond = condition on license. Produces counts: `n_inspections`, `n_investigations`, `n_enforcement`, `n_civil_fines`, `n_stop_placement`, `n_conditions`, `n_docs_total`, plus `latest_year` and `latest_enforcement_year`. This is the source of the **enforcement-intensity** measures.

A4. **WA_AFH_3County_Enriched.csv** — Master analysis dataset (primary file)

- **Rows x cols:** 3,457 × 41 (this is the file to use for replication)
- **Source:** A2 (roster) joined to A3 (document tallies) on `License_Number`, plus derived analytic columns.
- **Analysis performed (the core of the study):**
- **Operator-network clustering.** Built a normalized `phone_key` and `addr_key`; computed `phone_shared_count` = number of licenses sharing the same phone. Facilities with `phone_shared_count` ≥ 2 are flagged as belonging to a multi-license operator cluster.
- **Exclusion cross-check.** Tokenized facility names (dropping generic tokens like "ADULT FAMILY HOME / LLC / CARE") and matched distinctive tokens against the WA HCA/DSHS termination-and-exclusion list. Result: **0 verified name matches** in the three counties (reported transparently as a null finding).
- **Enforcement flags.** `has_enforcement`, `has_stop_placement`, `has_condition` booleans from A3 counts.
- **Headline results reproduced from this file:** | Metric | King | Pierce | Spokane | Pooled | |---|---|---|---|---| | Facilities | 1,795 | 1,052 | 610 | 3,457 | | In operator clusters | 95 | 47 | 22 | 164 | | **Clustering rate** | **5.3%** | **4.5%** | **3.6%** | **4.7%** | | Enforcement rate | 4.5% | 5.8% | 3.8% | — |
- Operator clusters total: **72**; facilities in clusters: **164**; facilities with ≥1 enforcement action: **165**.

A5. **WA_AFH_Facility_RedFlags.csv** — Facility-level review-priority indicator coding

- **Rows x cols:** 3,457 × 24
- **Source:** Derived from A4 by applying the MN-derived RF-01–RF-20 indicator typology at the facility level.
- **Analysis performed:** Coded the *per-facility computable* review-priority indicators only:
- `RF03_multi_home_operator` = `phone_shared_count` ≥ 2
- `RF08_network_cluster` = `phone_shared_count` ≥ 3
- `high_complaint_load` = `n_investigations` ≥ 3
- `RF06_exclusion_match` = exclusion-list name hit (0 in pilot)
- plus `has_enforcement` / `has_stop_placement` / `has_condition`. Indicators that require non-public claims data (e.g., billing concentration) are **not** coded here and are documented as non-testable in A8.

A6. **WA_Operator_PhoneClusters.csv** — The 72 operator clusters

- **Rows x cols:** 72 × 8
- **Source:** Aggregation of A4 over `phone_key` for clusters of ≥2 licenses.
- **Analysis performed:** For each cluster: `n_licenses`, member facility names, counties, cities, and summed `total_enforcement` / `total_investigations` / `total_civil_fines`. This is the operator-concentration backbone of Section 6.

A7. **WA_County_Concentration.csv** — County summary table

- **Rows x cols:** 3 × 12 (one row per county)
- **Source:** `groupby(County)` aggregation of A4.
- **Analysis performed:** Per county: facilities, beds, avg beds, count of facilities with enforcement / investigations / stop-placement / conditions, total civil fines, `pct_of_3county_facilities`, `enforcement_rate_pct`, `investigation_rate_pct`.

A8. **WA_RedFlag_Testability_Matrix.csv** — RF-01–RF-20 testability map

- **Rows x cols:** 20 × 6
- **Source:** Analyst-constructed mapping of each MN-derived indicator to WA public-data availability.
- **Analysis performed:** For each RF: the WA data source, whether it is **publicly testable** (Yes / Partial / No), the WA signal/metric, and status notes. This document defines the *scope and limits* of what public data can support.

A9. **WA_County_Normalized_Risk.csv** — Population-normalized risk

- **Rows x cols:** 9 × 21
- **Source:** A7 joined to U.S. Census **ACS** population & 65+ counts (via Census Reporter).
- **Analysis performed:** Computed `AFHs_per_1000_seniors`, `beds_per_1000_seniors`, `enforcement_actions_per_1000_seniors`, etc., so counties of different sizes are comparable.

Carries `confidence_level` labels.

A10. `WA_AFH_Growth_2013_2026.csv` — Growth time series

- **Rows x cols:** 14 × 8 (one row per year 2013–2026)
- **Source:** Official DSHS/BERK figures (2013–2018, 2022, 2024) + DSHS ArcGIS current roster (2026); intervening years **interpolated**.
- **Analysis performed:** Assembled a longitudinal active-AFH count. **Every reconstructed year carries an explicit `confidence_level` of "Low" and a note saying it is interpolated, not an official count.** Official anchors are labeled "High". Companion chart: `WA_AFH_Growth_2013_2026.png`.

A11. `WA_Medicaid_AFH_Contract_Trends_HISTORICAL.csv`

- **Rows x cols:** 4 × 10
 - **Source:** DSHS BERK 2018 report, Tables 3.1/3.4 (official).
 - **Analysis performed:** Tracks the share of AFHs holding a Medicaid contract (~88% across 2013–2018), establishing AFHs as a predominantly Medicaid-funded sector.
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PART B — MINNESOTA DATASETS (`mn_data/`)

B1. `MN_Medicaid_Master_Evidence_Base.xlsx` — Master workbook (primary file)

- **Tabs:** README, High-Risk Categories (14), Enforcement Cases (22), Revalidation Data, Program Growth.
- **Source:** DOJ, IRS-CI, MN Attorney General / MFCU press releases; MN DHS; CMS FY2023 PI review; OLA audits. Each row carries a source attribution.
- **Analysis performed:** Consolidation of all MN enforcement and program-integrity evidence into one workbook; basis for the RF-01–RF-20 framework and the MN figures.

B2. `MN_Medicaid_Fraud_Cases_Dataset.csv` — Charged/alleged cases

- **Rows x cols:** 22 × 8
- **Fields:** `Defendant`, `Entity`, `Program` (HSS/PCA/ICS/IHS/NEMT/ARMHS/HCBS), `Amount`, `Charges`, `Status`, `Source`, `Red_Flags`.
- **Analysis performed:** Each case tagged with the indicator pattern(s) it exhibits (RF-03, RF-09, RF-10, etc.), mapping real enforcement actions to the typology. **All entries labeled by charge status ("Charged (date)", "Under Investigation"); none asserted as adjudicated guilt.**

B3. `MN_Medicaid_Program_Growth_Data.csv`

- **Rows x cols:** 12 × 4 (`Program`, `Year`, `Spending`, `Source`)
- **Analysis performed:** Program-spending growth series (e.g., Housing Stabilization Services) underlying the MN growth chart (`HSS_Spending_Growth_Chart.png`) and the RF-01/RF-02 rapid-growth argument.

B4. `MN_Medicaid_Revalidation_Data.csv`

- **Rows x cols:** 10 × 3 (`Metric`, `Value`, `Source`)
 - **Analysis performed:** Tabulates MN DHS "Revalidate 2026" provider-revalidation outcomes used to discuss enrollment-integrity screening.
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PART C — SUPPORTING DOCUMENTS

Bundled under `wa_docs/` and `mn_docs/`: source bibliographies (with URLs + access dates), the data-availability matrices, the WA↔MN comparison matrix, the readiness memo, the review-priority indicator library (Minnesota source typology), and PDFs of each. The 5 manuscript figures are under `figures/`.

PART D — ANALYSIS SCRIPTS (`scripts/`)

Every number is reproducible. Run order: 1. `build_afh_master.py` → A1 (pull ArcGIS roster) 2.

build_3county.py → A2 (county filter) 3. scrape_reports.py → A3 (RCS document scrape) 4.
build_enriched.py → A4 (join + clustering + exclusion check + county table) 5. build_clusters.py → A6
(operator clusters) 6. build_rf_coding.py → A5, A8 (review-priority indicator coding + testability) 7.
build_normalized.py → A9 (Census normalization) 8. build_growth.py / build_growth_chart.py →
A10/A11 + chart 9. MN: build_deliverables.py, build_excel.py → B1–B4 10. build_charts.py,
manuscript/make_figures.py → figures

Confidence & limitations (read before citing)

- **Official, high-confidence:** current AFH roster, per-facility public-document counts, county splits, operator-cluster counts, enforcement counts.
- **Reconstructed/estimated (Low confidence, labeled in-file):** 2019–2021 and 2025 growth-year interpolations.
- **Proxy measures:** operator "clusters" are inferred from *shared phone numbers*, a public proxy for common operation — not a legal finding of common ownership.
- **Null finding (reported honestly):** 0 exclusion-list name matches among the 3,457 pilot facilities.
- Public data cannot test billing-level indicators (no public per-provider Medicaid payment data); those are flagged "No/Partial" in A8.